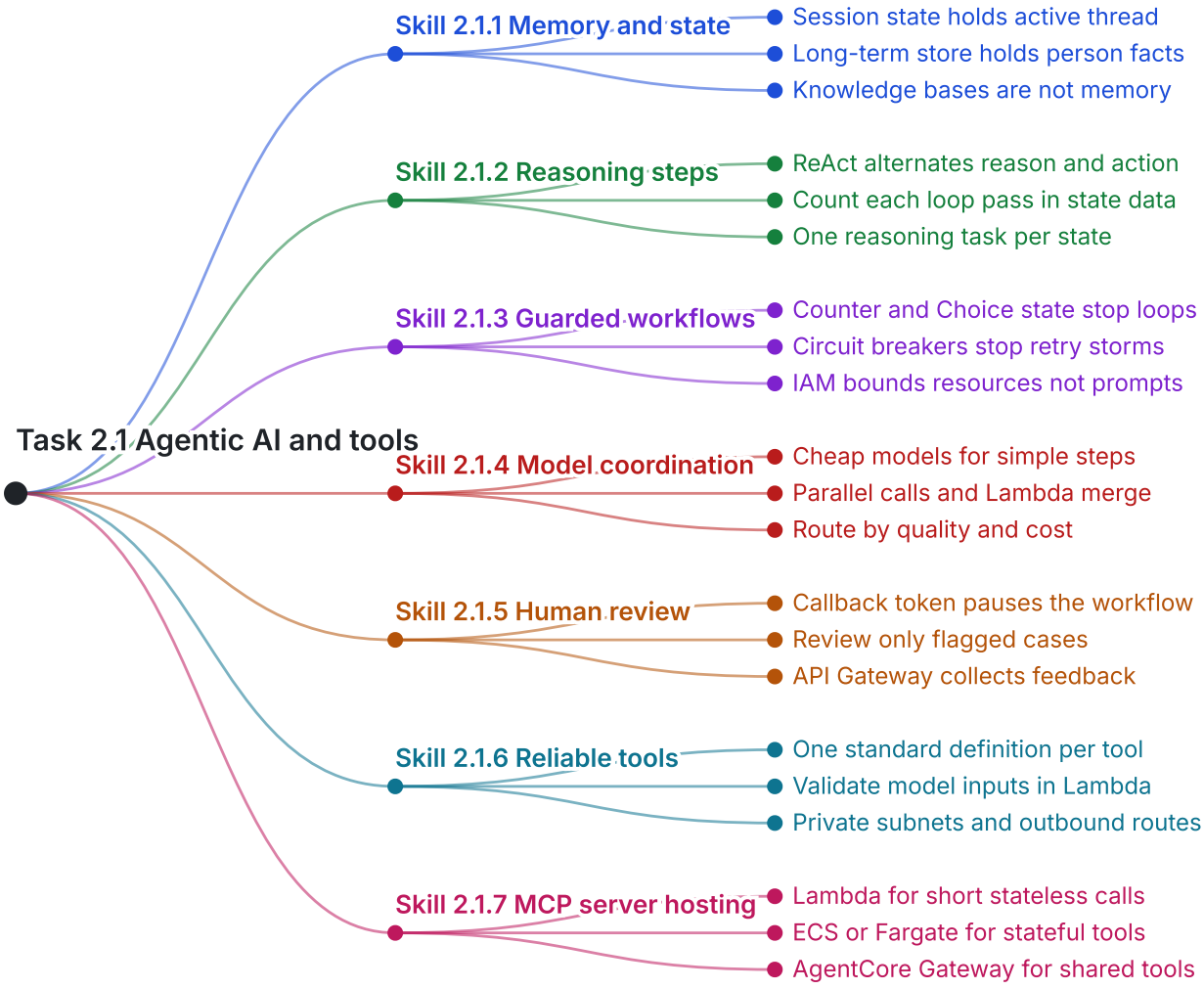
Memory sheet: Task 2.1 Agentic AI solutions and tool integrations

Domain 2, weight 26 percent. Read this sheet, then answer the flash questions aloud.



Deciding rules

When the stem says	Think of	Not when
Every step can be drawn before the request arrives	Workflow where code picks each step	The model must choose the next step
The model must decide which tool to call	Strands agent loop that owns its tools	Requests must route to one of several separate agents
One classifier must send each request to an agent	AWS Agent Squad with an intent classifier	One agent runs tool calls inside its own loop
Many clients or frameworks must call the same tools	MCP tools behind a shared gateway	Only one framework needs one local tool
A reviewer may reply after several days	Standard workflow with a callback token	The work is a short burst under five minutes
A loop could repeat after tool errors	Counter in state data and a Fail state	The fix is a prompt rule or a longer Lambda timeout
Every new execution calls a partner that is down	Circuit breaker with a shared status record	A single execution hits one short fault
A fact must return at the next visit for one person	Long-term memory record scoped to that person	The fact lasts for this session only, or is a document

Etch lines

1. Pick code when every step is known before the request arrives.
2. Strands runs one agent loop with tools chosen by the model.
3. Agent Squad routes requests, but does not replace a tool loop.
4. Read the verb in the question first. Decide or choose points to a model.
5. Standard runs up to one year and Express up to five minutes.
6. Put resource limits in IAM and per-call checks in a policy outside the prompt.

Flash questions

- Q: When is an agent loop the wrong design for a multi-step task? A: When every step is known before the request, so code should own the steps.
- Q: Which Step Functions type fits an approval that may take several days? A: Standard, with a callback token and HeartbeatSeconds set.
- Q: What stops a loop of model and tool calls from running until a cost limit ends it? A: A counter in state data with a Choice state that routes to a Fail state.
- Q: Where should a preference go if it must return at the next visit? A: A long-term record scoped to one person, not session state or a knowledge base.
- Q: Which pattern stops many executions from repeating calls to one failing service? A: A circuit breaker with a shared status record kept outside each execution.
- Q: Why does raising a Lambda timeout fail to stop a repeating loop? A: The loop keeps calling the model and tools, so cost grows and a counter is still needed.

- Q: Which host fits a stateless MCP tool whose calls are short? A: Lambda, with any state kept in a table or cache outside the function.
- Q: When should an MCP server run on ECS or Fargate instead of Lambda? A: When it keeps state, loads a large engine, or runs longer than Lambda allows.

Source: Task 2.1 deciding constraint, Skills 2.1.1 to 2.1.7, Confusable pairs in scope, and Read a long question. AWS docs and the study guide sections above, no practice-test text.